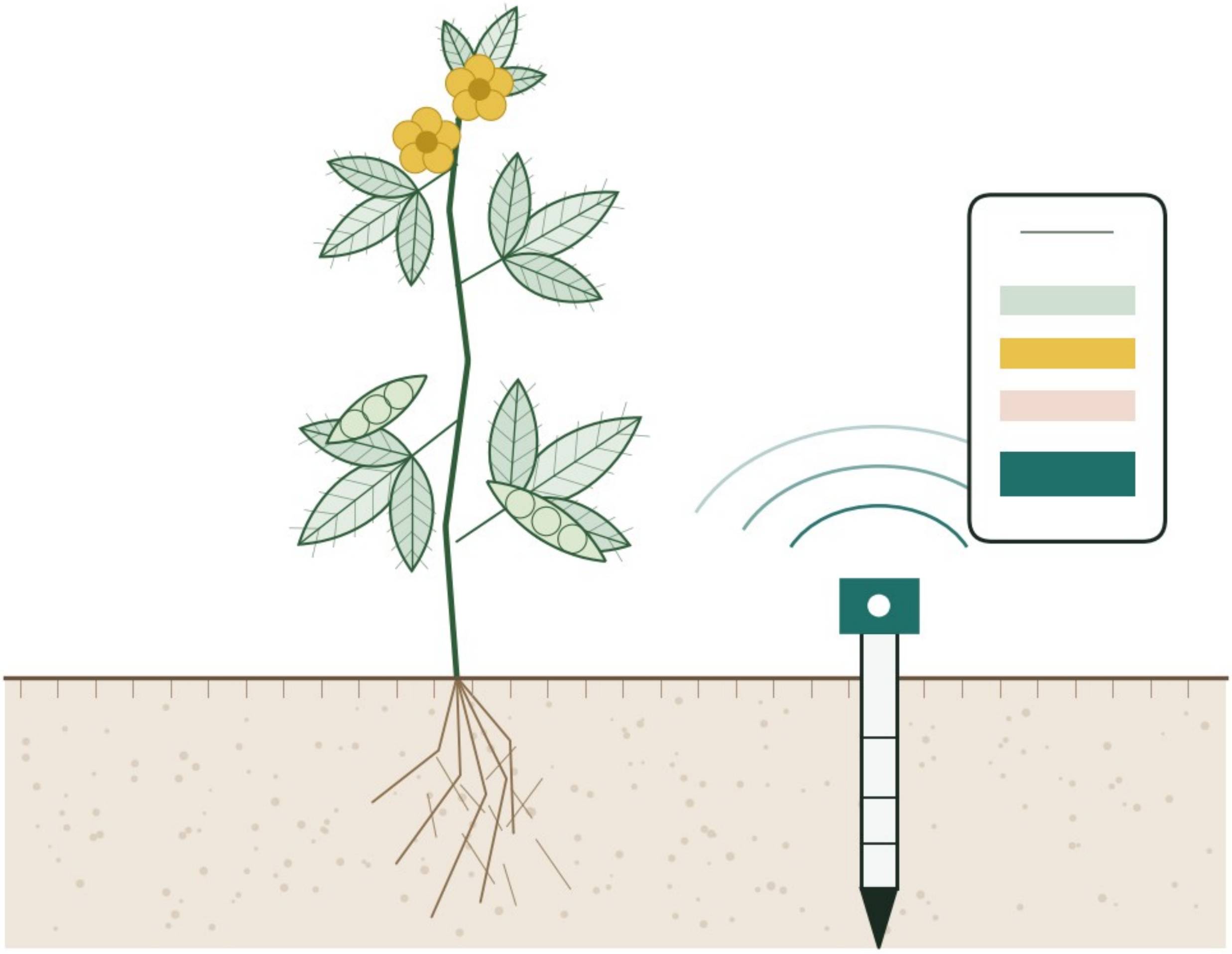
GROUP O

Software Requirements Specification

Multi-Site Smart Soil and Environmental Monitoring System



TEMPERATURE · MOISTURE · EC · pH · NITROGEN · PHOSPHORUS · POTASSIUM

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Document Control

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Stakeholder and end user	Ms Goitsemang Dikane
Prepared by	Group O project delivery team
Source repository	https://github.com/Nido-Maphosa86/soil-sensor-monitoring
Shared backend	https://supabase.com/dashboard/project/dsejhpcdshrkpmaewbcq

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Revision History

Version	Date	Summary of change
1.0	25 August 2026	Initial specification prepared from the stakeholder requirements document and the consultation meeting of 19 August 2026
2.0	30 August 2026	Revised following the stakeholder follow-up responses. Data provisioning changed to representative sample data; user roles fixed at two; threshold basis established; feasibility, risk register, change log and refined scope incorporated

Version	Date	Summary of change
3.0	4 September 2026	Findings and requirements confirmed by the stakeholder on 3 September 2026. Repository and backend details added, and Appendices A to E incorporated. Issued as the approved baseline for design and development.

Contents

1. Introduction

1.1 Purpose of this Document

This specification defines what the Multi-Site Smart Soil and Environmental Monitoring System must do, the conditions under which it must operate, and how its components work together. It is the agreed reference against which the system is designed, built and evaluated.

The document is written for two audiences. The stakeholder uses it to confirm that the system being built addresses the problem described. The delivery team uses it to guide design, divide development work and prepare test cases.

1.2 Scope of this Document

The specification covers the complete prototype: a mobile application, a web application, a shared backend, and the data that flows between them. It records the requirements agreed with the stakeholder, the assessment of whether those requirements can be delivered within the available time and resources, the risks identified, and the scope decisions that follow from both.

It does not cover the sensor hardware, its installation, calibration or maintenance. That equipment is owned and operated by a separate party.

1.3 Document Structure

The document follows the order in which the work was carried out. Sections 2 to 4 describe the stakeholder, the current situation and what the system is intended to achieve. Sections 5 to 9 define the boundaries of the system and how it is put together. Sections 10 to 16 state the requirements. Sections 17 to 20 record the feasibility assessment, the risks, and the resulting changes to scope. Section 21 lists the supporting evidence.

1.4 Basis of this Specification

Every requirement in this document traces to a statement made by the stakeholder, either in the written requirements document supplied by the organisation or in the written responses to the delivery team's follow-up questions. Both are held as supporting evidence and are listed in Section 21. The stakeholder confirmed these findings and approved the requirements on 3 September 2026.

1.5 How to Read This Document

A specification uses a small number of terms in a particular way. They are explained here so that the rest of the document can be read without a technical background. A fuller list of terms appears in Section 6.

1.5.1 The Four Types of Requirement

Requirements are separated into four kinds. Each answers a different question about the system, and each is numbered with its own prefix so that any requirement can be referred to later by its identifier.

Type	Question it answers	Example
Functional FR	What must the system do?	A user must be able to see the battery level of a sensor.
Non-functional NFR	How well, or under what conditions, must it do it?	Only an administrator may change an acceptable range.
System SR	What must the system be built	The mobile application must run on Android.

Type	Question it answers	Example
	with, or run on?	
Integration IR	How must the separate parts work together?	A user account created on the website must also work in the mobile application.

The distinction between the first two matters in practice. “A user must be able to view alerts” describes a function the system performs. “Only an administrator may change an acceptable range” describes a rule that governs how that function behaves. Keeping them apart makes it clear what will be built, and separately what it will be judged against.

1.5.2 Priority

Not every requirement carries the same weight. Each is given one of three priorities, so that if time becomes short it is already agreed what is set aside first. The priorities follow the order the stakeholder gave when asked which functions matter most, recorded in Section 14.2.

Priority	Meaning
Must Have	Essential. The system does not address the problem without it, and it will be built first.
Should Have	Important, and intended, but the system would still be usable without it in this first version.
Could Have	Included only if time and resources allow once everything above is complete.

1.5.3 Reading a Requirements Table

Each requirement is listed on one row with four columns.

Column	What it means
ID	A permanent reference, such as FR39. Requirement numbers do not change, even if a requirement is later altered or set aside, so that any discussion can point to a specific line.
Requirement	A single statement of one thing the system must do or satisfy. Each is written to be testable, so that at the end it can be shown to be either met or not met.
Component	Which part of the system it applies to. Mobile is the phone application; Web is the website; Both means it applies to each; Backend is the shared part behind them, which stores the information and applies the rules, and which users do not see directly.
Priority	Must Have, Should Have or Could Have, as described above.

1.5.4 What the Numbering Means

Requirement identifiers are permanent. If a requirement is simplified or deferred, it keeps its number and the change is written into the change log in Section 19 rather than the requirement being deleted. This means the record always shows what was originally asked for, what was agreed instead, and why. Nothing disappears quietly.

2. Stakeholder and Current Process

2.1 Stakeholder Profile

Item	Detail
Organisation	Department of Agriculture, Plant Sciences, Faculty of Health and Environmental Sciences, Central University of Technology, Free State.
Activities	Plant science research and the monitoring of soil and environmental conditions for agricultural benefit.
Stakeholder and end user	Ms Goitsemang Dikane, Plant Sciences (nGAP), Department of Agriculture. Confirms the requirements and accepts the system on behalf of the intended users.
Engagement channel	Ms Mpho Mbele, Departmental Manager: Information Technology, through whom the written requirements document was supplied and the follow-up questions were routed.
Deployment	Fifteen 7-in-1 RS485 soil sensors installed across three agricultural locations, five sensors at each location.
Crop	Pigeon pea (Cajanus cajan) at all three locations.
Reason for engagement	The organisation has sensing equipment in place and measurements being produced, but no single system through which those measurements can be viewed, compared and acted upon.

2.2 Current Process

Soil condition at the three locations is currently assessed by physical sampling. Soil samples are taken once, at the start of the planting cycle, and sent to a laboratory. The laboratory result is used to determine which fertiliser is suitable for the season.

Between that single test and the end of the season, the organisation has no continuous view of soil condition. Fifteen sensors are deployed and producing measurements, but there is no system that brings those readings together, compares conditions across the three locations, or indicates when a site requires attention. Assessment between sampling points therefore depends on physical inspection of each location.

2.2.1 Participants in the Current Process

Participant	Involvement
Site or farm manager	Responsible for the condition of the land and for decisions about intervention
Agricultural staff	Carry out work at the locations based on those decisions
Field workers	Present at the locations and first to observe visible changes
Researchers	Interpret conditions and assess results across the three sites
Laboratory	Analyses the soil samples taken at the start of the planting cycle
Sensor operator	Owns and operates the sensing equipment and publishes the measurements

2.2.2 Current Process Flow

Figure 1 shows the same process in diagram form.

Step	Action	Frequency
1	Soil samples are taken at each location	Once, at the start of the planting cycle
2	Samples are sent to a laboratory for analysis	Once per cycle
3	Laboratory results determine the fertiliser to be applied	Once per cycle
4	Sensors measure soil and environmental conditions continuously	Every 30 minutes
5	Measurements are published by the sensor operator	Continuous
6	Conditions between sampling points are assessed by physical inspection	Irregular
7	Changes in condition are identified when noticed on site	Irregular, and often late

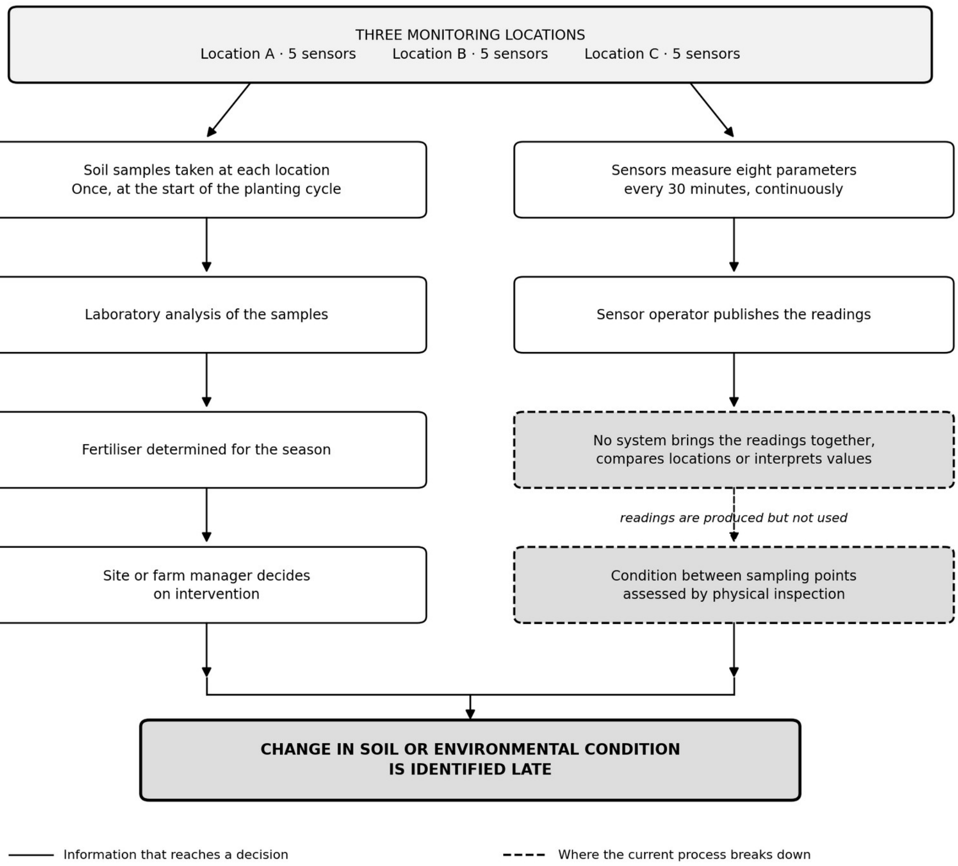


Figure 1 — Current process at the three monitoring locations

2.3 Difficulties in the Current Process

Difficulty	Effect on the organisation
Soil condition is tested only once per cycle	Conditions change during the season with no corresponding measurement being reviewed

Difficulty	Effect on the organisation
Readings are produced but not brought together	Fifteen sensors generate data continuously that no one is in a position to read
No comparison across the three locations	A site performing differently from the others is not identified
Change in condition is noticed late	Responses to low soil moisture, unusual nutrient levels or other environmental change are delayed
Assessment depends on physical presence	Three separate locations cannot be inspected frequently enough to be useful
No visibility of equipment condition	A sensor that has stopped reporting is indistinguishable from a sensor reporting normal conditions

3. Problem Statement

The organisation currently uses single-point laboratory soil sampling and physical inspection to assess soil condition across three agricultural locations. This results in significant changes in soil and environmental conditions being identified late, while measurements capable of revealing those changes are produced continuously by fifteen deployed sensors and are not accessible in any usable form.

The problem affects site and farm managers, agricultural staff, field workers and researchers, because decisions about intervention depend on information they cannot see. Conditions such as low soil moisture or unusual nutrient levels are noticed only when they become visible on site or when the next sampling cycle occurs, by which time a response is late.

An improved solution is required to bring the readings from all three locations into one place, interpret them against acceptable ranges for the crop grown, indicate which location or sensor requires attention, report whether the sensing equipment is functioning, and make that information available to the responsible people without requiring a visit to the site.

3.1 Elements of the Problem

Element	Statement
Current situation	Soil is tested once per planting cycle. Sensor measurements are produced continuously but are not presented in a usable form.
Problem	Significant change in soil and environmental condition is identified late.
Affected users	Site or farm managers, agricultural staff, field workers and researchers.
Consequences	Delayed response to low soil moisture, unusual nutrient levels and other environmental change. Equipment failure goes unnoticed.
Need	A central system that organises the readings, shows the condition of each site, and identifies changes requiring attention.

4. Project Aim and Objectives

4.1 Aim

To deliver an integrated prototype that brings soil and environmental measurements from fifteen sensors across three agricultural locations into a single system, interprets those measurements against acceptable ranges for the crop, reports the operational condition of the sensing equipment, and presents the result to authorised users through a mobile and a web application.

4.2 Objectives

1. Store soil and environmental readings for all fifteen sensors in a single shared database, without duplication.
2. Present the current condition of each location and each individual sensor.
3. Compare measured values against thresholds configured for each location and raise an alert when a value falls outside its acceptable range.
4. Report the operational condition of each sensor, including battery level, sensor status and time of last communication.
5. Raise an alert when a sensor has stopped communicating.
6. Retain readings over time and present them as graphs for a user-selected period.
7. Restrict access to information and configuration according to the user's assigned role.
8. Allow an administrator to manage locations, sensors, thresholds and user accounts.
9. Allow conditions at the three locations to be compared side by side.
10. Notify responsible users when a location requires attention.

5. Project Scope

Scope records what the project includes and what it does not. Agreeing this early prevents a common difficulty, where each side assumes something different was included and only discovers the difference at the end.

5.1 Deliverables

Deliverable	Description
Mobile application	Flutter application for viewing conditions, sensor detail, equipment status, history and alerts
Web application	ASP.NET MVC application for administration, configuration, comparison and reporting
Shared backend	Supabase project providing the database, authentication, access rules and storage
Data provisioning component	Loads representative sample readings into the database at a defined interval
Software Requirements Specification	This document, maintained as the agreed reference
Project schedule	Baselined schedule showing tasks, responsibilities, dependencies and milestones
Source repository	Version-controlled repository containing documentation and source code
Integrated demonstration	Working demonstration of the prototype against the acceptance criteria

5.2 Acceptance Criteria

These are the conditions the finished system must demonstrate before it is accepted. Each is written so that it can be shown to be either met or not met, rather than being a matter of judgement. Together they are the practical test of whether the problem described in Section 3 has been addressed.

Ref	Criterion
AC1	An authorised user can sign in, and an unauthorised user cannot.
AC2	A dashboard displays counts of locations, registered sensors, active sensors, inactive sensors and open alerts.
AC3	All three locations are listed, and the sensors at each location can be reached from the location.
AC4	All fifteen sensors are registered and individually identifiable.
AC5	All seven measured parameters are displayed for a selected sensor, together with the time of the reading.
AC6	Battery level, sensor status and time of last communication are displayed for each sensor.
AC7	A reading falling outside its configured threshold generates an alert visible in both applications.
AC8	A sensor that has not communicated within the configured period is reported as inactive and generates an alert.
AC9	Readings for a selected sensor and period are displayed as a graph.

Ref	Criterion
AC10	An administrator can create, edit and remove locations, sensors, thresholds and user accounts.
AC11	A standard user cannot change any configuration.
AC12	Conditions at the three locations can be compared in a single view.
AC13	Data recorded through the web application is visible in the mobile application, and the reverse where applicable.
AC14	Readings can be exported from the web application.

5.3 Exclusions

- Installation, configuration, calibration or maintenance of the sensor hardware
- Direct communication between the applications and the sensors or gateway equipment
- Live integration with the sensor operator's published data store, which is treated as a later integration step
- Agronomic advice, fertiliser recommendation or crop treatment guidance beyond comparison against configured thresholds
- Automated irrigation or any control of physical equipment
- Predictive analysis or forecasting of future soil condition
- Payment processing
- Public access to readings without an authenticated account

5.4 Constraints

- The prototype must be delivered within one academic semester, with approximately three weeks available for development
- The technology set is fixed: Flutter for mobile, ASP.NET MVC for web, Supabase for the shared backend
- Live sensor readings are produced by a separate party and are not available to the delivery team for this version
- Services must operate within free service tiers
- Eight team members with differing availability and levels of development experience
- Only representative sample data may be used in the prototype and in the demonstration

5.5 Assumptions

- The three locations, the fifteen sensors and the seven measured parameters remain as described
- Pigeon pea remains the crop at all three locations
- Two user roles are sufficient for this version
- The stakeholder remains available to confirm findings and to evaluate the prototype
- Sample data structured as described in Section 8.3 is an acceptable substitute for live readings
- The free service tier is sufficient for the volume of readings the prototype generates

6. Definitions and Abbreviations

6.1 Terms Used About the System

Term	Meaning
Alert	A record raised automatically when a reading falls outside its configured threshold, or when a sensor stops communicating
ASP.NET MVC	The framework used to build the web application
Dashboard	The summary view presenting counts and overall condition across all locations
EC	Electrical conductivity, a measure of dissolved salts in the soil, expressed in decisiemens per metre
Flutter	The framework used to build the mobile application
Inactive sensor	A sensor that has not produced a reading within the configured period
Location	One of the three sites at which sensors are deployed
NPK	Nitrogen, phosphorus and potassium
Parameter	One of the eight quantities measured by a sensor
Reading	One complete set of measurements captured by one sensor at one point in time
Role	The permission level assigned to a user account, either Administrator or Standard User
RS485	The serial communication standard used by the sensor equipment
Sensor	One of the fifteen deployed measuring devices
Sensor status	The reported operational state of a sensor
Supabase	The shared backend providing the database, authentication, access rules and storage
Threshold	A configured acceptable range for one parameter at one location

6.2 Terms Used About the Project

These terms describe how the work is planned and recorded rather than what the system does.

Term	Meaning
Acceptance criterion	A condition the finished system must demonstrate before it is accepted. The full list is in Section 5.2. Each is written so that it can be shown to be either met or not met, rather than judged as a matter of opinion.
Assumption	Something taken to be true for planning purposes, which has not been separately verified. If an assumption turns out to be wrong, the plan may need to change.
Constraint	A limit the project must work within, such as available time or the technologies that must be used. A constraint is not a choice.
Contingency	What will be done if a risk actually happens.

Term	Meaning
Deliverable	Something the delivery team will produce and hand over.
Entity	A kind of information the system stores, such as a location or a reading. Section 16 lists them.
Exclusion	Something deliberately not included, recorded so that expectations are clear on both sides.
Feasibility	Whether the project can realistically be completed with the time, skills and resources available. Section 17 assesses this in six areas, each reaching one of three conclusions: feasible, feasible with changes, or not currently feasible.
Impact	How seriously a risk would affect the project if it happened, rated from 1 to 3.
Issue	A problem that has already happened. This differs from a risk, which has not.
Likelihood	How likely a risk is to happen, rated from 1 to 3.
Mitigation	What will be done in advance to make a risk less likely, or less damaging if it occurs.
Prototype	A working version built to demonstrate the proposed solution. It shows that the approach works, and is not a finished product for daily operational use.
Risk	Something that may happen and would affect the project if it did. A risk that has already happened is no longer a risk; it is an issue.
Risk score	Likelihood multiplied by impact, giving a number from 1 to 9. Scores of 1 to 3 are low, 4 to 6 are medium, and 7 to 9 are high.
Scope	What the project includes and, equally importantly, what it does not.
Specification	This document. It records what has been agreed, and is the reference the finished system is measured against.

7. Stakeholders and User Groups

7.1 Stakeholders

Stakeholder	Interest in the system
Department of Agriculture, Plant Sciences	Commissions the system. Represented by Ms Goitsemang Dikane, who confirms the requirements and accepts the prototype as the end user.
Sensor operator	Owns and operates the sensing equipment and publishes the measurements. Not a user of the system.
Delivery team	Designs, builds and delivers the prototype

7.2 User Roles

The stakeholder confirmed that two roles are required for this version, and that additional roles may be introduced later if needed.

Role	Permissions
Administrator	Full access. Manages user accounts and roles, locations, sensor registration and threshold configuration. Views all readings, history, alerts and reports.
Standard User	Read access. Views the dashboard, locations, sensors, current readings, equipment status, history and alerts. Cannot change any configuration.

7.3 User Groups by Function

Group	Typical role	Application
Site or farm manager	Administrator or Standard User	Both
Agricultural staff	Standard User	Mobile
Field workers	Standard User	Mobile
Researchers	Standard User	Both
System administrator	Administrator	Web

The exact number of users is not fixed. The system does not impose a limit.

8. Product Perspective

8.1 System Composition

The prototype operates as a single integrated system built on one shared backend. The mobile application and the web application both connect to the same Supabase project and read from the same tables. Neither application holds its own copy of the data.

Component	Technology	Function
Mobile application	Flutter	Viewing conditions, sensor detail, equipment status, history and alerts in the field
Web application	ASP.NET MVC	Administration, configuration, comparison, reporting and export
Shared backend	Supabase	Database, authentication, role-based access rules, storage
Data provisioning	Scheduled backend function	Loads representative sample readings at a defined interval

8.2 Data Ownership Boundaries

Each table has exactly one writer. This prevents two sources of truth for the same information.

Data	Written by	Read by
Readings	Data provisioning component only	Both applications
Sensor status	Data provisioning component only	Both applications
Alerts	Backend only, generated automatically	Both applications
Locations	Web application only	Both applications
Sensors	Web application only	Both applications
Thresholds	Web application only	Both applications and the backend
Users and roles	Web application only	Backend access rules

The mobile application does not create or modify readings, configuration or user accounts. Alerts are generated in the backend so that both applications present the same alert state.

8.3 Data Provisioning

For this version the system operates on representative sample data rather than live sensor readings. The stakeholder confirmed in writing that this is acceptable, that access to the operator's published data store is not required at this stage, and that live integration may be treated as a later step once access and the final data format are available.

Sample readings are generated to match the structure and behaviour of real readings: fifteen sensors across three locations, a complete reading every thirty minutes, values within and occasionally outside the configured thresholds, and sensors that occasionally stop reporting so that inactivity detection can be demonstrated.

8.3.1 Reading Record Structure

Field	Type	Description
Sensor identifier	Reference	The sensor that produced the reading
Location identifier	Reference	The location at which the sensor is installed
Timestamp	Date and time	When the reading was captured
Soil moisture	Numeric	Percentage
Soil temperature	Numeric	Degrees Celsius
Electrical conductivity	Numeric	Decisiemens per metre
pH	Numeric	pH units
Nitrogen	Numeric	Milligrams per kilogram
Phosphorus	Numeric	Milligrams per kilogram
Potassium	Numeric	Milligrams per kilogram

8.3.2 Sensor Status Record Structure

Field	Type	Description
Sensor identifier	Reference	The sensor being reported on
Battery level	Numeric	Percentage of remaining charge
Power status	Text	Reported power or charging state
Sensor status	Text	Active or inactive
Last communication	Date and time	When the sensor last produced a reading
Recorded at	Date and time	When the status record was written

8.3.3 Later Integration

The data provisioning component is the only part of the system that writes readings. When live integration is required, that component is replaced with one that retrieves readings from the operator's published data store and maps them to the same structure. No other part of the system changes. This isolation is the reason the component exists as a separate element rather than as sample data loaded directly into the database.

9. Main Product Functions

Component	Functions
Mobile application	Sign in and out; view the dashboard summary; view the three locations; view the sensors at a location; view all seven parameters for a selected sensor; view battery level, power status and sensor status; view time of last communication; view readings over time as graphs; view open alerts and alert detail; compare conditions across locations.
Web application	Sign in and out; view the dashboard summary; create, edit and remove locations; register, edit and remove sensors; create and manage user accounts and roles; configure thresholds per location and parameter; view and filter historical readings; compare locations; view and manage alerts; export readings and alerts.
Shared backend	Store users, roles, locations, sensors, readings, sensor status, thresholds and alerts; authenticate users; enforce role-based access; evaluate readings against thresholds; detect inactive sensors; generate alerts; retain historical readings.
Data provisioning	Generate and store representative readings for all fifteen sensors at a thirty-minute interval; record sensor status; avoid duplicate records.

10. Functional Requirements

Functional requirements state what the system must do. Each is written as a single action that can be shown to work or not work, and each is numbered so that it can be referred to later. They are grouped by area of the system, and the grouping follows the order of the stakeholder priority list in Section 14.2.

The four columns are explained in Section 1.5.3, and the three priority levels in Section 1.5.2.

10.1 Authentication and Access

ID	Requirement	Component	Priority
FR01	A user must be able to sign in using an email address and password.	Both	Must
FR02	The system must reject sign-in attempts with invalid credentials.	Both	Must
FR03	A user must be able to sign out.	Both	Must
FR04	The system must assign each user account one of two roles: Administrator or Standard User.	Backend	Must
FR05	The system must restrict every function according to the signed-in user's role.	Backend	Must
FR06	The system must prevent access to any reading, alert or configuration by an unauthenticated request.	Backend	Must
FR07	A user must be able to change their own password.	Both	Should

10.2 Dashboard

ID	Requirement	Component	Priority
FR08	The system must display a dashboard on sign-in showing the overall state of the deployment.	Both	Must
FR09	The dashboard must display the number of monitoring locations.	Both	Must
FR10	The dashboard must display the number of registered sensors.	Both	Must
FR11	The dashboard must display the number of active sensors.	Both	Must
FR12	The dashboard must display the number of inactive sensors.	Both	Must
FR13	The dashboard must display the number of open alerts.	Both	Must
FR14	The dashboard must indicate the overall condition of each location.	Both	Must
FR15	The dashboard must display the time at which the information was last updated.	Both	Should

10.3 Locations

ID	Requirement	Component	Priority
FR16	The system must list all monitoring locations.	Both	Must
FR17	A user must be able to open a location and view the sensors installed at it.	Both	Must
FR18	The system must display the current condition of a selected location.	Both	Must
FR19	The system must display the number of open alerts at a selected location.	Both	Must
FR20	An administrator must be able to create a location.	Web	Must
FR21	An administrator must be able to edit the details of a location.	Web	Must
FR22	An administrator must be able to remove a location that has no sensors assigned to it.	Web	Should

10.4 Sensors

ID	Requirement	Component	Priority
FR23	The system must list all sensors installed at a selected location.	Both	Must
FR24	Each sensor must be individually identifiable.	Both	Must
FR25	A user must be able to open a sensor and view its detail.	Both	Must
FR26	The system must display which sensor produced any given reading.	Both	Must
FR27	An administrator must be able to register a sensor and assign it to a location.	Web	Must
FR28	An administrator must be able to edit the details of a registered sensor.	Web	Must
FR29	An administrator must be able to deactivate a sensor without deleting its readings.	Web	Should
FR30	An administrator must be able to move a sensor to a different location.	Web	Could

10.5 Readings

ID	Requirement	Component	Priority
FR31	The system must display the most recent reading for a selected sensor.	Both	Must
FR32	The system must display soil moisture for a selected sensor.	Both	Must
FR33	The system must display soil temperature for a selected sensor.	Both	Must
FR34	The system must display electrical conductivity for a selected sensor.	Both	Must
FR35	The system must display pH for a selected sensor.	Both	Must
FR36	The system must display nitrogen, phosphorus and potassium for a selected sensor.	Both	Must
FR37	The system must display the date and time at which each reading was captured.	Both	Must
FR38	The system must indicate, for each displayed parameter, whether the value falls inside or outside its configured threshold.	Both	Must

10.6 Equipment Condition

ID	Requirement	Component	Priority
FR39	The system must display the battery level of each sensor.	Both	Must
FR40	The system must display the power status of each sensor.	Both	Must
FR41	The system must display whether each sensor is active or inactive.	Both	Must
FR42	The system must display the time at which each sensor last communicated.	Both	Must
FR43	The system must mark a sensor as inactive when it has not produced a reading within the configured period.	Backend	Must
FR44	An administrator must be able to configure the period after which a sensor is treated as inactive.	Web	Should

10.7 Thresholds

ID	Requirement	Component	Priority
FR45	The system must hold a configurable acceptable range for each parameter at each location.	Backend	Must
FR46	An administrator must be able to set the minimum and maximum value of a threshold.	Web	Must
FR47	An administrator must be able to edit an existing threshold.	Web	Must
FR48	The system must record the source or basis on which a threshold was set.	Web	Should
FR49	A standard user must be able to view thresholds but not change them.	Both	Must
FR50	An administrator must be able to apply one location's thresholds to another location.	Web	Could

10.8 Alerts

ID	Requirement	Component	Priority
FR51	The system must generate an alert when a reading falls outside its configured threshold.	Backend	Must
FR52	The system must generate an alert when a sensor becomes inactive.	Backend	Must
FR53	An alert must record the sensor, the location, the parameter, the measured value and the time raised.	Backend	Must
FR54	The system must list all open alerts.	Both	Must
FR55	A user must be able to open an alert and view its detail.	Both	Must
FR56	The system must present the same alert state in both applications.	Backend	Must
FR57	A user must be able to acknowledge an alert so that it is removed from the open list.	Both	Should
FR58	The system must notify users when a new alert is raised.	Both	Should
FR59	The system must filter alerts by location, by sensor and by severity.	Web	Should

10.9 History and Comparison

ID	Requirement	Component	Priority
FR60	The system must retain readings so that previous values can be viewed.	Backend	Must

ID	Requirement	Component	Priority
FR61	A user must be able to view previous readings for a selected sensor.	Both	Must
FR62	The system must display readings over time as a graph.	Both	Must
FR63	A user must be able to select the period shown in a graph.	Both	Must
FR64	A user must be able to select which parameter is graphed.	Both	Must
FR65	The system must allow conditions at the three locations to be compared in a single view.	Both	Must
FR66	The system must allow historical readings to be filtered by location, sensor, parameter and date range.	Web	Should

10.10 Reporting

ID	Requirement	Component	Priority
FR67	An administrator must be able to export readings for a selected location and period.	Web	Must
FR68	An administrator must be able to export the alert history for a selected period.	Web	Should
FR69	The system must produce a summary report of conditions across the three locations.	Web	Should

10.11 User Administration

ID	Requirement	Component	Priority
FR70	An administrator must be able to create a user account.	Web	Must
FR71	An administrator must be able to assign a role to a user account.	Web	Must
FR72	An administrator must be able to change a user's role.	Web	Must
FR73	An administrator must be able to deactivate a user account.	Web	Must
FR74	The system must list all user accounts and their assigned roles.	Web	Must

10.12 Data Provisioning

ID	Requirement	Component	Priority
FR75	The system must store readings for all fifteen sensors at a thirty-minute interval.	Backend	Must
FR76	The system must store a sensor status record for each sensor.	Backend	Must
FR77	The system must not store duplicate readings for the same sensor and timestamp.	Backend	Must
FR78	The system must record the outcome of each provisioning run so that failures are visible.	Backend	Should

11. Non-Functional Requirements

Non-functional requirements describe qualities, rules and conditions rather than functions. They answer how well the system must work and under what rules, rather than what it does. A requirement such as “only an administrator may change an acceptable range” does not add a feature; it governs how an existing feature behaves.

These are grouped below by the kind of condition they describe.

11.1 Security and Access

ID	Requirement	Priority
NFR01	Passwords must be stored in a form from which the original cannot be recovered.	Must
NFR02	Access rules must be enforced in the backend and not only in the user interface.	Must
NFR03	Backend keys and service credentials must be held outside the source repository.	Must
NFR04	Credentials must not be embedded in the mobile application.	Must
NFR05	Only representative sample data and test accounts may be used in the prototype and in the demonstration.	Must
NFR06	The system must record which account performed each configuration change.	Should

11.2 Usability

ID	Requirement	Priority
NFR07	Every screen must be reachable within three interactions from the dashboard.	Must
NFR08	A parameter outside its threshold must be visually distinguishable from one inside it.	Must
NFR09	Error messages must state what went wrong and what the user can do about it.	Must
NFR10	The mobile interface must display correctly on an Android smartphone.	Must
NFR11	The system must confirm to the user when an action has succeeded or failed.	Must
NFR12	Screen labels and terminology must be consistent across both applications.	Should

11.3 Performance and Reliability

ID	Requirement	Priority
NFR13	The dashboard must display without unnecessary delay for the current data volume.	Must
NFR14	A graph covering thirty days for one sensor must render without	Must

ID	Requirement	Priority
	unnecessary delay.	
NFR15	The failure of a provisioning run must not prevent existing readings from being viewed.	Must
NFR16	Readings must be retained for at least twelve months.	Should

11.4 Correctness and Boundaries

ID	Requirement	Priority
NFR17	The system must compare measured values against configured thresholds and must not present the result as agronomic or treatment advice.	Must
NFR18	Every displayed reading must show the time at which it was captured, so that the age of the information is visible.	Must
NFR19	A threshold must record the basis on which it was set, so that its origin can be established.	Should
NFR20	Functions not completed in the prototype must be documented rather than presented as working.	Must

12. System Requirements

System requirements state what the system must be built with and what it must run on. They are recorded because they constrain the design, and because they were fixed before the project began rather than chosen by the delivery team.

ID	Requirement
SR01	The mobile application must be developed using Flutter.
SR02	The web application must be developed using ASP.NET MVC.
SR03	The shared backend must be provided by Supabase.
SR04	The database must be relational and must enforce referential integrity between locations, sensors, readings, thresholds and alerts.
SR05	Authentication must be provided by the shared backend rather than implemented separately in each application.
SR06	Role-based access must be enforced by database access rules.
SR07	Source code and project documentation must be held in a version-controlled repository.
SR08	The mobile application must run on Android.

13. Integration Requirements

The system is made of separate parts: a phone application, a website and a shared store of information behind both. Integration requirements state how those parts must work together, so that they behave as one system rather than three that happen to look similar. These are the requirements most easily overlooked, and the ones most likely to cause difficulty if they are.

ID	Requirement
IR01	Both applications must connect to the same Supabase project and read from the same tables.
IR02	Configuration recorded through the web application must be visible in the mobile application.
IR03	Alerts generated in the backend must be visible in both applications with the same state.
IR04	Readings written by the data provisioning component must be available to both applications.
IR05	A user account created through the web application must be able to sign in to the mobile application.
IR06	A threshold changed through the web application must be applied to subsequent alert evaluation without further action.
IR07	The data provisioning component must be the only writer of readings and sensor status, so that it can later be replaced by a live integration without changing any other component.

14. Requirements Prioritisation

14.1 Summary

Type	Prefix	Total	Must	Should	Could
Functional	FR	78	61	14	3
Non-functional	NFR	20	16	4	0
System	SR	8	8	0	0
Integration	IR	7	7	0	0
Total	—	**113**	**92**	**18**	**3**

14.2 Stakeholder Priority Order

The stakeholder was asked which functions matter most if development time becomes limited. The order below is the answer given, and it governs the sequence in which work is completed.

Order	Function	Requirements
1	User login and access control	FR01–FR06
2	The three monitoring locations	FR16–FR19
3	Display of all fifteen sensors	FR23–FR26
4	Current sensor readings	FR31–FR37
5	Battery and sensor status	FR38–FR42
6	Historical readings	FR59–FR64
7	Alerts for abnormal conditions and inactive sensors	FR51–FR56
8	A clear dashboard for each location	FR08–FR14

The stakeholder identified advanced analytics, prediction and complex reporting as suitable for a later version. Those are not included in this specification.

15. Agricultural Threshold Basis

Thresholds are configurable per location and per parameter. The values below are the initial settings, established from the South African Department of Agriculture production guideline for pigeon pea and from standard soil salinity classification. They are a starting point for configuration, not fixed limits, and the stakeholder may adjust any of them.

Parameter	Minimum	Maximum	Basis
pH	5.0	7.0	Pigeon pea production guideline. Excessive acidity inhibits nodulation and causes chlorosis and die-back.
Soil temperature	18 °C	35 °C	Pigeon pea production guideline growing range. See note below.
Soil moisture	20 %	60 %	Growth-stage dependent. See note below.
Electrical conductivity	0	2 dS/m	Standard salinity classification. Above 2 dS/m is slightly saline; above 4 dS/m is saline.
Nitrogen	Configurable	Configurable	Pigeon pea fixes atmospheric nitrogen and shows little response to nitrogen fertiliser. Configured against laboratory soil analysis.
Phosphorus	Configurable	Configurable	Configured against laboratory soil analysis.
Potassium	Configurable	Configurable	The guideline notes that adequate potassium is required. Configured against laboratory soil analysis.

15.1 Stated Limitations of the Initial Values

11. The production guideline gives air temperature, not soil temperature. The soil temperature range above is an approximation applied from that figure and should be reviewed against local observation.
12. Soil moisture requirements vary by growth stage. Pigeon pea prefers moist conditions during its first two growing months and drier conditions during flowering and harvesting. A single fixed range will produce alerts during flowering that do not indicate a problem. Stage-aware thresholds are recorded as a later enhancement.
13. Nitrogen, phosphorus and potassium have no single acceptable range independent of the soil. These are configured from the laboratory analysis the organisation already obtains at the start of each planting cycle.
14. The system reports whether a measured value falls inside or outside a configured range. It does not recommend fertiliser, irrigation or any other intervention.

15.2 Agreed Position on Calibration

The stakeholder reviewed these values and confirmed the approach in writing on 3 September 2026, noting that the calibration and setting of range measurements can be amended once the platform has been established, and that the specification meets the entry requirements of the end user as it stands.

This settles a question the delivery team could not answer alone. The initial values above are a working starting point rather than final limits. Because every range is stored in the backend and configured per location, refining them later requires no change to the applications and no new development. The pipeline will be revisited with the stakeholder once the system is running and real readings can be compared against the ranges in use.

16. Minimum Database Entities

An entity is a kind of information the system stores. A location is one entity, a reading is another. Listing them here establishes what the system must keep a record of. How they relate to one another, and the exact structure of each, is settled during design.

The final structure and relationships are produced during system design. The entities below are the minimum required to satisfy the requirements in this document.

Entity	Purpose	Key attributes
Users	System users and assigned roles	Identifier, email, role, account status, created date
Locations	The three monitoring sites	Identifier, name, description, created date
Sensors	The fifteen registered devices	Identifier, name, location, model, installation date, active status
Readings	Individual measurements with timestamps	Identifier, sensor, timestamp, moisture, temperature, conductivity, pH, nitrogen, phosphorus, potassium
Sensor status	Operational condition of each device	Identifier, sensor, battery level, power status, sensor status, last communication, recorded at
Thresholds	Acceptable ranges per parameter and location	Identifier, location, parameter, minimum, maximum, basis, updated by, updated at
Alerts	Automatically generated warnings	Identifier, sensor, location, alert type, parameter, measured value, severity, state, raised at, acknowledged by, acknowledged at

17. Feasibility Study

The assessment below covers whether the system as specified can be delivered within the available time, skills and resources.

17.1 Technical

The application layer uses frameworks the delivery team has worked with previously. The required functions are conventional application development: authenticated access, list and detail screens, form-driven management, filtered queries, charting and export.

The uncertainty that existed at the start of the project concerned retrieving data from a system operated by a separate party. The stakeholder has since confirmed that live integration is not required for this version and that representative sample data is acceptable. That removes the dependency the delivery team did not control.

Skills still to be developed are database access rules, scheduled backend functions, and charting in both frameworks. Time for this is allocated before development begins.

Decision	Feasible
Reason	The required technologies are within the team's demonstrated capability, and the external dependency that carried the technical risk has been removed from this version.

17.2 Operational

The system requires no change to how measurements are produced and no manual data entry by users. Users read information rather than supplying it, which keeps the burden of adoption low.

Two roles were confirmed by the stakeholder as sufficient. Configuration functions are restricted to administrators, because a threshold changed by one user alters the alerts every other user sees.

Guidance required is minimal for the mobile application, which is read-only. Brief guidance is needed for threshold configuration and user management on the web application.

Decision	Feasible
Reason	The solution fits the organisation's existing activities, the user roles are confirmed, and the functions map directly to the difficulties described in Section 2.3.

17.3 Economic

This assessment covers whether the delivery team has access to the resources required, rather than a business financial analysis.

Resource	Position
Development tools	Flutter, the .NET SDK, the development environments and version control are available at no cost.
Shared backend	Supabase operates within its free tier at the volume this prototype generates.

Resource	Position
Hardware procurement	None. The sensing equipment is owned and operated by a separate party.
Development machines	Confirmed available across the team.
Test devices	At least one Android device is required for the demonstration.
Internet access	Required by both applications and by the shared backend.
Presentation resources	Venue, display and a reliable connection for the demonstration.

Decision	Feasible
Reason	All required software operates within free tiers, no hardware procurement falls to the delivery team, and the resources needed are available.

17.4 Schedule

Approximately three weeks are available for development across a mobile application, a web application and a shared backend. The delivery team has eight members, which permits parallel work, but components developed separately require time to combine and that time is planned rather than assumed.

Removing the live integration from this version reduces the work by one component and removes the item that could not be designed until an external data format was known.

Required changes

15. Development is planned backwards from the acceptance criteria. Work not contributing to one of them is deferred.
16. The database and access rules are completed before parallel application development begins, because both applications depend on them.
17. Time for combining components is reserved within the development window rather than added at the end.
18. Each of the mobile, web, backend and provisioning components has a named owner.

Decision	**Feasible with changes**
Reason	The available time is sufficient only if scope is restricted to the acceptance criteria and the shared backend is completed before parallel development begins.

17.5 Legal

Area	Position
Personal information	The system stores user accounts only: email address, assigned role and account status. Measurements are not personal information.
Data protection	Handling of personal information is subject to the Protection of Personal Information Act. Restricting the prototype to test accounts keeps the project outside the handling of real personal information.
Data ownership	Measurements produced by the sensing equipment belong to the operating party. This version does not access them.
Credentials	Backend keys and passwords are held outside the source repository and are not distributed in client applications.
Confidentiality	Stakeholder correspondence is held as evidence. Contact details and identifiers are excluded from the source repository.
Attribution	External sources used in this document are cited where used.

Decision	**Feasible**
Reason	The prototype handles no real personal information, does not access third-party data in this version, and the confidentiality request has been applied.

17.6 Ethical

Area	Position
Purpose understood	The stakeholder understands that this is a prototype with a defined delivery period, not a production system.
Information collected	Only the account details required for authentication and role assignment.
Risk of over-reliance	If the system presented threshold comparison as agronomic advice, a user might act on a recommendation the system is not qualified to make. The system reports measured values against configured ranges and offers no crop or treatment recommendation. This is recorded as NFR17.
Use of information	Measurements are used to display conditions, raise alerts and produce reports for the organisation, and for no other purpose.
Fair treatment	Access is assigned by role rather than by individual. Incomplete functions are documented rather than presented as working.

Decision	Feasible
Reason	No unnecessary information is collected, information is used only for its stated purpose, and an explicit boundary is recorded against advice the system is not qualified to give.

17.7 Summary

Area	Decision	Main reason	Change required
Technical	Feasible	Required technologies within the team's capability; external dependency removed from this version	None
Operational	Feasible	Users read information rather than entering it; roles confirmed	Configuration restricted to administrators
Economic	Feasible	Software free or already held; no hardware procurement	None
Schedule	Feasible with changes	Approximately three development weeks across three components	Restrict scope to the acceptance criteria; complete the backend before parallel work
Legal	Feasible	No real personal information; no third-party data accessed in this version	None
Ethical	Feasible	Explicit boundary against agronomic advice	None

17.8 Conclusion

The proposed project is feasible with changes.

The main strengths of the project are that the required technologies are within the delivery team's demonstrated capability, that the software and services needed are available at no cost, that no hardware procurement or maintenance falls to the delivery team, and that the system requires no change to how the organisation currently produces its measurements. Users read information rather than supplying it, which keeps the burden of adoption low.

The main limitation identified is time. Approximately three weeks are available for development across a mobile application, a web application and a shared backend, and eight people working in parallel carry an integration cost that must be planned rather than assumed.

To ensure that the project can be completed within the available period, the delivery team will plan development backwards from the acceptance criteria and defer work that does not contribute to one of them, complete the database and access rules before any parallel application development begins, reserve time for combining components within the development window, assign a named owner to each component, and defer the additional features listed in Section 20 until the required functions are complete.

18. Project Risk Register

A risk is something that may happen and would affect the project if it did. It is recorded before it happens, so that action can be taken in advance rather than in response. Something that has already happened is not a risk but an issue, and is handled separately.

Each risk below is rated for how likely it is and how serious it would be. Multiplying the two gives a score that allows risks to be compared and ranked. Each also carries a mitigation, meaning what will be done in advance to reduce it, and a contingency, meaning what will be done if it happens anyway. Every risk has one named person responsible for watching it.

18.1 Rating Scale

Rating	Likelihood	Impact
1	Unlikely to occur	Minor delay or inconvenience
2	Possible	Noticeable effect requiring action
3	Likely	Serious effect on delivery or quality

Risk score = likelihood × impact. Scores of 1 to 3 are Low, 4 to 6 are Medium, and 7 to 9 are High.

18.2 Register

ID	Risk	Category	L	I	Score	Level
R01	Development time may be insufficient for the full scope, leaving required functions incomplete	Schedule	3	3	9	High
R02	Components developed in parallel may fail to integrate, so that parts work individually but not together	Technical	2	3	6	Medium
R03	Credentials may be committed to the repository, exposing keys that deletion alone cannot withdraw	Security	2	3	6	Medium
R04	Threshold values may prove unsuitable for the sites, producing alerts that do not indicate a real problem	Quality	2	3	6	Medium
R05	Contribution may be uneven across an eight-member team, concentrating work on a few people	Team	2	2	4	Medium
R06	Scope may expand during development as features are added mid-build	Scope	2	2	4	Medium
R07	A team member may become unavailable, leaving a component without an owner	Team	2	2	4	Medium
R08	The stakeholder may be unavailable when confirmation or evaluation is required	Stakeholder	2	2	4	Medium
R09	Sample data may not resemble real readings closely enough to demonstrate	Quality	2	2	4	Medium

ID	Risk	Category	L	I	Score	Level
	the system convincingly					
R10	Learning time for access rules and scheduled functions may exceed the allowance made	Technical	2	2	4	Medium
R11	Free service limits may be reached as readings accumulate	Resource	1	2	2	Low
R12	Later live integration may reveal a data structure the current design cannot accommodate	Technical	2	1	2	Low

18.3 Response and Ownership

ID	Mitigation	Contingency	Owner
R01	Plan development backwards from the acceptance criteria; complete the backend before parallel work; reserve integration time explicitly	Deliver Must Have requirements only; document incomplete functions rather than presenting partial features as complete	Project manager
R02	Finalise the database structure before application development begins; agree field names and types in writing; integrate early rather than at the end	Reserve time in the final week for correction; prioritise the integrated demonstration over individual features	Technical lead
R03	Commit an ignore file before any configuration exists; hold keys in environment variables; brief every member before their first commit	Rotate the exposed key immediately. The file must be removed and the credential replaced, as deletion does not remove it from history	Technical lead
R04	Record the basis of every threshold; make all values configurable; state the known limitations in Section 15.1	Adjust values with the stakeholder and record the change in the change log	Specification team
R05	Assign named responsibility for every task; record contributions with evidence; review progress at a fixed weekly meeting	Reallocate outstanding work and record the reallocation	Project manager
R06	Record every requirement with a priority; treat Should Have and Could Have as excluded from the initial build; log any change with a reason	Return added work to the backlog and document the decision	Specification team
R07	Ensure at least two members understand each component; keep all work in the shared repository rather than on individual machines	Redistribute the affected work and record the change	Project manager
R08	Agree confirmation and evaluation dates in advance; put requests in writing	Proceed on previously confirmed findings and request written feedback	Project manager
R09	Generate values within realistic ranges, including occasional threshold breaches and periods of sensor inactivity	Adjust the generator; demonstrate against a prepared data set	Backend developer
R10	Allocate explicit learning time before development begins; build a small working example of each first	Simplify the affected function and document the simplification	Technical lead
R11	Monitor database size; store readings at the defined interval rather than at maximum frequency; use summarised data for long-range charts	Archive older readings; reduce the provisioning interval	Backend developer
R12	Isolate all provisioning in one component so that a replacement affects nothing else	Adapt the replacement component only; no other component changes	Technical lead

18.4 Highest Risks

R01 is the only High risk. R02, R03 and R04 share a score of 6 and are treated as the next priority.

Rank	Risk	Why it matters	Action now
1	R01 — Insufficient development time	Approximately three weeks are available for fourteen acceptance criteria across three components. Scope control is the only available response and it must be applied before development starts, not during it.	Treat the refined scope in Section 20 as closed. Sequence the backend first. Assign a named owner to each component.
2	R02 — Integration failure	The prototype is evaluated as one integrated system. Components that work individually but do not combine would not satisfy the integration requirements, and integration problems surface late unless deliberately tested early.	Agree the database structure and field names in writing before any application work begins. Build one end-to-end path early.
3	R04 — Unsuitable thresholds	Alerts are the point of the system. Thresholds that fire when nothing is wrong, or stay silent when something is, would undermine the whole prototype in front of the stakeholder.	Confirm the initial values in Section 15 with the stakeholder. Keep every value configurable and record its basis.

18.5 Monitoring

The register is reviewed at each weekly progress meeting. For each open risk the delivery team records whether likelihood or impact has changed, whether the mitigation has been carried out, and whether the risk can be closed. New risks are added with the same fields. A risk that materialises is recorded as an issue, together with the action taken.

19. Requirements Change Log

A specification changes as understanding improves. Rather than quietly editing the document, every change is recorded here with the reason for it, so that the history of what was agreed remains visible to both sides.

A requirement that has been set aside is never simply deleted. It is recorded below, with the reason, so that it can be revisited in a later version without having to be rediscovered.

Ref	Original position	Change	Reason
C01	Readings retrieved from the sensor operator's published data store and written to the shared backend	Replaced by a data provisioning component using representative sample data	The stakeholder confirmed in writing that live integration is not required for this version and that sample data is acceptable
C02	Live data retrieval listed as a project deliverable	Recorded as a later integration step	Follows from C01
C03	Battery and device status dependent on fields being published by the operator	Included as a confirmed requirement	The stakeholder confirmed that battery level, sensor status and last communication time must be present in the sample data
C04	An eighth parameter, humidity, had been recorded in an earlier draft	Removed. The specification records seven measured parameters	The stakeholder requirements document, section 3, defines seven soil parameters, and the sample reading fields supplied in the follow-up responses list the same seven. Humidity was not a stakeholder requirement
C05	User roles undefined	Fixed at two roles: Administrator and Standard User	The stakeholder confirmed two roles are sufficient for this version, with more added later if required
C06	Thresholds required but no values available	Initial values established from the pigeon pea production guideline and salinity classification, all configurable	The stakeholder confirmed the crop as pigeon pea and did not specify values
C07	Sensor inactivity undefined	A sensor is inactive after two hours without a reading, configurable	Reading interval confirmed at thirty minutes; two hours represents four consecutive missed readings
C08	Reading frequency unspecified	Set at thirty minutes	Confirmed by the stakeholder
C09	Notification method unspecified	Push notification recorded as the requirement, delivered as in-application alerts in the first version	Confirmed as the preferred channel. Simplified for the first version to reduce integration work
C10	Complex reporting included	Reduced to export of readings and alerts	The stakeholder identified advanced analytics, prediction and complex reporting as suitable for a later version
C11	Prediction and analytics	Removed	Identified by the stakeholder as a later

Ref	Original position	Change	Reason
	under consideration		version
C12	System to interpret readings and recommend action	Boundary recorded as NFR17	The system compares values against configured ranges and does not offer agronomic advice
C13	Historical retention unspecified	Set at twelve months	Proposed by the delivery team; recorded as NFR16
C14	Threshold values researched by the delivery team and not yet agreed	Confirmed as a working starting point, to be refined once the system is running	The stakeholder confirmed on 3 September 2026 that calibration and range settings can be amended after the platform is established

20. Refined Project Scope

Following the feasibility assessment in Section 17 and the risks in Section 18, every proposed function was reviewed against the time available. Each is marked with one of four decisions.

Decision	Meaning
Keep	Included as originally planned
Simplify	Included, but in a reduced form for this version, with the fuller version to follow
Postpone	Not built in this version, but recorded as a later enhancement rather than abandoned
Remove	Excluded, because it does not belong in this system

20.1 Required Functions

Function	Decision	Reason
Authentication and role-based access	Keep	First in the stakeholder's priority order and a dependency of every other function
Dashboard summary	Keep	Acceptance criterion AC2
Locations and sensor navigation	Keep	Acceptance criteria AC3 and AC4
Display of all seven parameters	Keep	The core purpose of the system
Battery, power and sensor status	Keep	Confirmed by the stakeholder as required
Inactive sensor detection	Keep	Named by the stakeholder in the priority order
Threshold configuration	Keep	Without it alerts have no basis
Alert generation and viewing	Keep	Acceptance criteria AC7 and AC8
Historical readings and graphs	Keep	Acceptance criterion AC9
Location comparison	Keep	Acceptance criterion AC12
User administration	Keep	Required for the organisation to operate the system independently
Export of readings	Keep	Acceptance criterion AC14
Data provisioning	Keep	Without it the system holds no data

20.2 Additional Functions

Function	Decision	Reason
Push notification	Simplify	Delivered as in-application alerts in the first version. Background push added once the required functions are complete.
Live dashboard updates	Simplify	Refresh on view rather than a continuous

Function	Decision	Reason
		subscription, which reduces integration work
Alert acknowledgement	Keep	Low cost, and without it the alert list becomes unusable as alerts accumulate
Alert filtering	Keep	Required once alert volume grows across fifteen sensors
Complex reporting and analytics	Remove	Identified by the stakeholder as suitable for a later version
Prediction and forecasting	Remove	Identified by the stakeholder as suitable for a later version
Offline mode	Postpone	The most time-intensive item proposed. Reconsidered if connectivity at the sites proves poor.
Map view of locations	Postpone	Adds a mapping dependency without contributing to an acceptance criterion
Sensor selection by code scanning	Postpone	Selection from a list achieves the same outcome
Biometric sign-in	Postpone	Standard authentication satisfies the access requirement
Site visit logging with photographs	Postpone	Introduces file storage and a write path from an application that currently reads only
Weather overlay	Postpone	Adds an external dependency without contributing to an acceptance criterion
Dark mode and animated transitions	Postpone	Presentation only
Home screen widget	Postpone	Platform-specific work outside the required scope
Interface language selection	Postpone	Retained as a later enhancement. Translation requires verification by a first-language speaker and is not undertaken until the required functions are complete.

21. Supporting Evidence

Appendix	Evidence	Status
A	Follow-up questions issued to the stakeholder	Held
B	Stakeholder written responses to the follow-up questions	Held
C	Stakeholder requirements document supplied by the organisation	Held
D	Stakeholder written confirmation of the findings, Ms Goitsemang Dikane, 3 September 2026	Held
E	Engagement record: meeting dates, attendees and correspondence	Held

Contact details, identity numbers, credentials and confidential records are excluded from the appendices and from the source repository.

21.1 Confirmation

A written summary of the findings was issued to the stakeholder for review. Ms Goitsemang Dikane confirmed the findings and the requirements recorded in this document on 3 September 2026, stating that the document meets the entry requirements of the end user and can be used as it stands.

She further noted that minor corrections can be amended as the work proceeds, and that the calibration and setting of range measurements can be revisited once the platform has been established. Those positions are recorded in Section 15.2 and in change log entry C14. The confirmation is held as Appendix D, and this version is issued as the approved baseline for design and development.

Confirmation	Detail
Reviewed by	Ms Goitsemang Dikane
Position	Plant Sciences (nGAP), Department of Agriculture — stakeholder and end user
Date of confirmation	3 September 2026
Findings confirmed as accurate	Yes
Requirements approved	Yes — the document meets the entry requirements of the end user
Corrections required	None blocking. Minor corrections to be amended as the work proceeds.
Outstanding for a later stage	Calibration and range settings to be revisited once the platform is established
Evidence	Written confirmation held as Appendix D

21.2 Sources

Reference	Source
S1	Department of Agriculture, Forestry and Fisheries, Republic of South Africa. Pigeon peas production guideline. Directorate: Plant Production.
S2	Food and Agriculture Organization of the United Nations. Crop salt tolerance data, salinity classification by electrical conductivity.
S3	Project requirements document, Multi-Site Smart Soil Monitoring System, supplied by the stakeholder organisation.
S4	Stakeholder written responses to the follow-up questions issued by the delivery team.

21.3 Project Repository and Backend

Project documentation and source code are held in a single version-controlled repository. The shared backend is a single Supabase project used by both applications.

Item	Detail
Source repository	https://github.com/Nido-Maphosa86/soil-sensor-monitoring
Repository name	soil-sensor-monitoring
Repository owner	Nido-Maphosa86
Repository contents	README, documentation folder holding this specification and its appendices, the project schedule, and the source code for the mobile application, the web application and the backend
Access	All eight members of the delivery team have write access
Supabase project	https://supabase.com/dashboard/project/dsejhpcdshrkpmaewbcq
Supabase project reference	dsejhpcdshrkpmaewbcq
Backend contents	PostgreSQL database, authentication, access rules and storage for readings, sensor status, thresholds and alerts

No password, API key, service role key, telephone number, identity number or confidential organisational record has been committed to the repository at any point. A .gitignore file was committed before any configuration file was created.

SUPPORTING EVIDENCE

Appendices A to E

Stakeholder engagement evidence for the Multi-Site Smart Soil and Environmental Monitoring System

Appendix	Evidence	Date
A	Follow-up questions issued to the stakeholder	26 August 2026
B	Stakeholder written responses to the follow-up questions	27 August 2026
C	Project requirements document supplied by the stakeholder	20 August 2026
D	Stakeholder written confirmation of the findings	3 September 2026
E	Engagement record — meetings, correspondence and attendance	19 Aug – 3 Sep 2026

Traceability

Every requirement in this specification traces to one of these sources. Section 2 draws on the meeting of 19 August and on Appendix B; sections 5 and 10 to 13 draw on Appendix C; section 14.2 draws on Appendix B; section 15 draws on the meeting of 19 August and on Appendix D.

Appendix A — Follow-Up Questions Issued

Issued	26 August 2026
From	Nido Maphosa, on behalf of Group O
To	Ms Mpho Mbele
Reason	A meeting could not be arranged during that week, so the questions were put in writing

The covering note read: "Thank you again for the time you have given to this project so far, and for the requirements document you shared with us. As we move into the planning and design we have a short list of follow-up questions that will help us complete our documentation accurately and design the system correctly."

#	Question
1	Before this monitoring system was proposed, how was soil condition on these three sites being assessed — was there a process in place at all, or is this a completely new capability for your operation?
2	What specifically prompted you to commission this system — was there a particular moment or realisation that made you decide you needed it?
3	Could you confirm, for our documentation, your organisation's full name, what your organisation does, and your role and title?
4	What crop or crops are grown at each of the three locations, and what soil type is typical at each?
5	Who specifically is affected day to day by not having this system — which roles or team members?
6	What are the practical consequences when this information isn't available in time — for example, crop stress, wasted water, or a delayed response?
7	Of the difficulties you've described, which one causes the most damage if left unaddressed?
8	Roughly how many people will use the system, and are there any user types beyond Administrator and Standard User that we should plan for?
9	Could you share a sample of how a single sensor reading looks once it is stored in Firebase — for example a screenshot or the field names — so that we can design our data integration correctly?
10	Does the data in Firebase include battery level, power status, and a last-communication timestamp for each sensor node?
11	What access will we be given to read the data in Firebase, and who is the right technical contact if the format ever changes?
12	If development time runs short, which functions matter most to you, and which could wait for a later version?
13	Beyond the technical requirements already defined, how would you personally know that this system has solved the problem for your operation?
14	Was anyone else involved in preparing the original requirements document, or should we consider you the sole author for our records?
15	Once we send you a written summary of what we've discussed, would you be able to confirm it is accurate by email?

Appendix B — Stakeholder Written Responses

Received	27 August 2026
From	Ms Mpho Mbele
Form	The same document returned with answers written beneath each question

Preamble supplied with the responses

"For the current prototype, you may use realistic dummy sensor data. Firebase integration is not required at this stage and can be considered later if needed."

Question 1

Before this monitoring system was proposed, how was soil condition on these three sites being assessed — was there a process in place at all, or is this a completely new capability for your operation?

Response	This is a new centralised monitoring capability for the project. At present, there is no single mobile and web system that continuously brings together soil and environmental readings from all three locations. The proposed system will make it easier to view and compare information from the different sites in one place. At the moment they only take soil samples and send them to the lab only once when they start planting just to see which fertilizer will be suitable.

Question 2

What specifically prompted you to commission this system — was there a particular moment or realisation that made you decide you needed it?

Response	The main reason for the system is the need to monitor conditions across three different locations without relying only on physical checks. The project will use 15 sensors, with five sensors at each location, so a central system is needed to organise the readings, show the condition of each site and help users identify changes that may need attention.

Question 3

Could you confirm, for our documentation, your organisation's full name, what your organisation does, and your role and title?

Response	This is for research purpose, you can call it environmental researchers, responsible for monitoring environment for agricultural benefit.

Question 4

What crop or crops are grown at each of the three locations, and what soil type is typical at each?

Response	The crop is pigeon pea, soil type is not a must to know. This does not prevent you from

	continuing with the system design and prototype development.

Question 5

Who specifically is affected day to day by not having this system — which roles or team members?

Response	Site or farm manager, agricultural staff, field workers, researchers.

Question 6

What are the practical consequences when this information isn't available in time — for example, crop stress, wasted water, or a delayed response?

Response	When the information is not available in time, changes in soil conditions may be noticed late. This can lead to delayed responses to problems such as low soil moisture, unusual nutrient levels or other environmental changes. The system should help users identify changes earlier and respond more quickly.

Question 7

Of the difficulties you've described, which one causes the most damage if left unaddressed?

Response	The most important problem to address is the delay in identifying significant changes in soil or environmental conditions across the three locations. A central monitoring system should make it easier to see which location or sensor needs attention.

Question 8

Roughly how many people will use the system, and are there any user types beyond Administrator and Standard User that we should plan for?

Response	The exact number of users is not fixed at this stage. For the first version, the system should support at least two user types: Administrator and Standard User. The Administrator can manage users, locations and sensor information, while the Standard User can view dashboards, readings, history and alerts. Additional roles can be added later if the stakeholder requires them.

Question 9

Could you share a sample of how a single sensor reading looks once it is stored in Firebase — for example a screenshot or the field names — so that we can design our data integration correctly?

Response	A dummy reading can include fields such as sensor ID, location ID, nitrogen (N), phosphorus (P), potassium (K), soil moisture, soil temperature, pH, electrical conductivity, battery level and date/time. The exact live-data structure can be integrated at a later stage if required.

Question 10

Does the data in Firebase include battery level, power status, and a last-communication timestamp for each sensor node?

Response	For the prototype, the dummy data should include battery level, sensor status and a last-communication date/time so that the system can demonstrate sensor-health monitoring. This is important because there will be 15 sensors across three locations. You do not need to obtain these values from Firebase at this stage.

Question 11

What access will we be given to read the data in Firebase, and who is the right technical contact if the format ever changes?

Response	No Firebase access is required for the current prototype. You can continue using dummy data and focus on the required system functionality, user interface, dashboards and monitoring features. Real sensor or database integration can be treated as a later integration step when access and the final data format are available.

Question 12

If development time runs short, which functions matter most to you, and which could wait for a later version?

Response	If development time becomes limited, priority should be given to user login and access control, the three monitoring locations, display of all 15 sensors, current sensor readings, battery and sensor status, historical readings, alerts for abnormal conditions or inactive sensors, and a clear dashboard for each location. More advanced analytics, prediction and complex reporting can be added in a later version.

Question 13

Beyond the technical requirements already defined, how would you personally know that this system has solved the problem for your operation?

Response	The system will be considered successful if authorised users can easily view the condition of all three locations from the mobile and web applications, identify which sensor produced a reading, notice unusual conditions, see whether a sensor is active or inactive, view previous readings and receive useful alerts. The aim is to make monitoring information easier to access, more organised and available sooner.

Question 14

Was anyone else involved in preparing the original requirements document, or should we consider you the sole author for our records?

Response	Consider me the sole author.

Question 15

Once we send you a written summary of what we've discussed, would you be able to confirm it is accurate by email?

Response	Yes.

Appendix C — Project Requirements Document

Received	20 August 2026
From	Ms Mpho Mbele, on behalf of the stakeholder organisation
Title	Project Requirements — Multi-Site Smart Soil Monitoring System
Form	PDF, thirty-one numbered sections

The document as received is reproduced after this summary. Its sections map to this specification as follows.

Section of the requirements document	Where it is addressed
1 Project overview	§4 Aim and objectives
2 Physical deployment — three locations, fifteen sensors	§2.1, §5.2 AC3 and AC4
3 Soil parameters — seven per sensor	§8.3.1, §10.5
4 User roles — Administrator and Standard User	§7.2
5 Authentication and access control	§10.1, §12
6 Location management	§10.3
7 Sensor management	§10.4
8 Sensor data collection	§8.3, §10.12
9 Sensor reading storage	§10.9, §16
10 Battery monitoring	§10.6
11 Device health monitoring	§10.6
12 Monitoring thresholds	§10.7, §15
13 Alerts	§10.8
14 Mobile application requirements	§9
15 Web application requirements	§9
16 Location comparison	FR65
17 Filtering	FR66
18 Reporting	§10.10
19 Data visualisation	§10.9
20 Supabase requirements	§8.1, §12
21 Minimum database entities	§16
22 Data validation	§10.12, NFR
23 System security	§11.1
24 Reliability	NFR15
25 Usability	§11.2, NFR8

Section of the requirements document	Where it is addressed
26 Performance	§11.3
27 Audit and timestamps	NFR18, §16
28 Testing requirements	Phase 3 planning
29 Required final demonstration	§5.2 Acceptance criteria
30 Integration requirement	§13, change log C01
31 Required technical components	§12, §5.3

Appendix D — Stakeholder Confirmation of the Findings

Received	3 September 2026
From	Ms Goitsemang Dikane, Plant Sciences (nGAP), Department of Agriculture
To	Ms Mpho Mbele and Nido Maphosa
Subject	Confirmation of findings — Multi-Site Smart Soil and Environmental Monitoring System
Outcome	Findings confirmed, requirements approved, no blocking corrections

The reply in full

"Dear Ms Mbele

Kindly note that I have gone through the document and have realized that some of the minor corrections can be amended while the process unfolds. The calibration and setting of range measurements can also be amended later on once the platform has been established. We shall revisit the pipeline and amend where necessary, for now I believe the document can be used as it meets the entry requirements as the end user.

Warm regards

Goitsemang Dikane, Plant Sciences: nGAP, Work Integrated Learning, Department of Agriculture, Faculty of Health and Environmental Sciences"

How this was applied

What she said	Where it is recorded
The document meets the entry requirements of the end user and can be used as it stands	§21.1 Confirmation
Minor corrections can be amended while the process unfolds	§21.1, recorded as non-blocking
Calibration and range settings can be amended once the platform is established	§15.2 and change log C14
The pipeline will be revisited and amended where necessary	§15.2

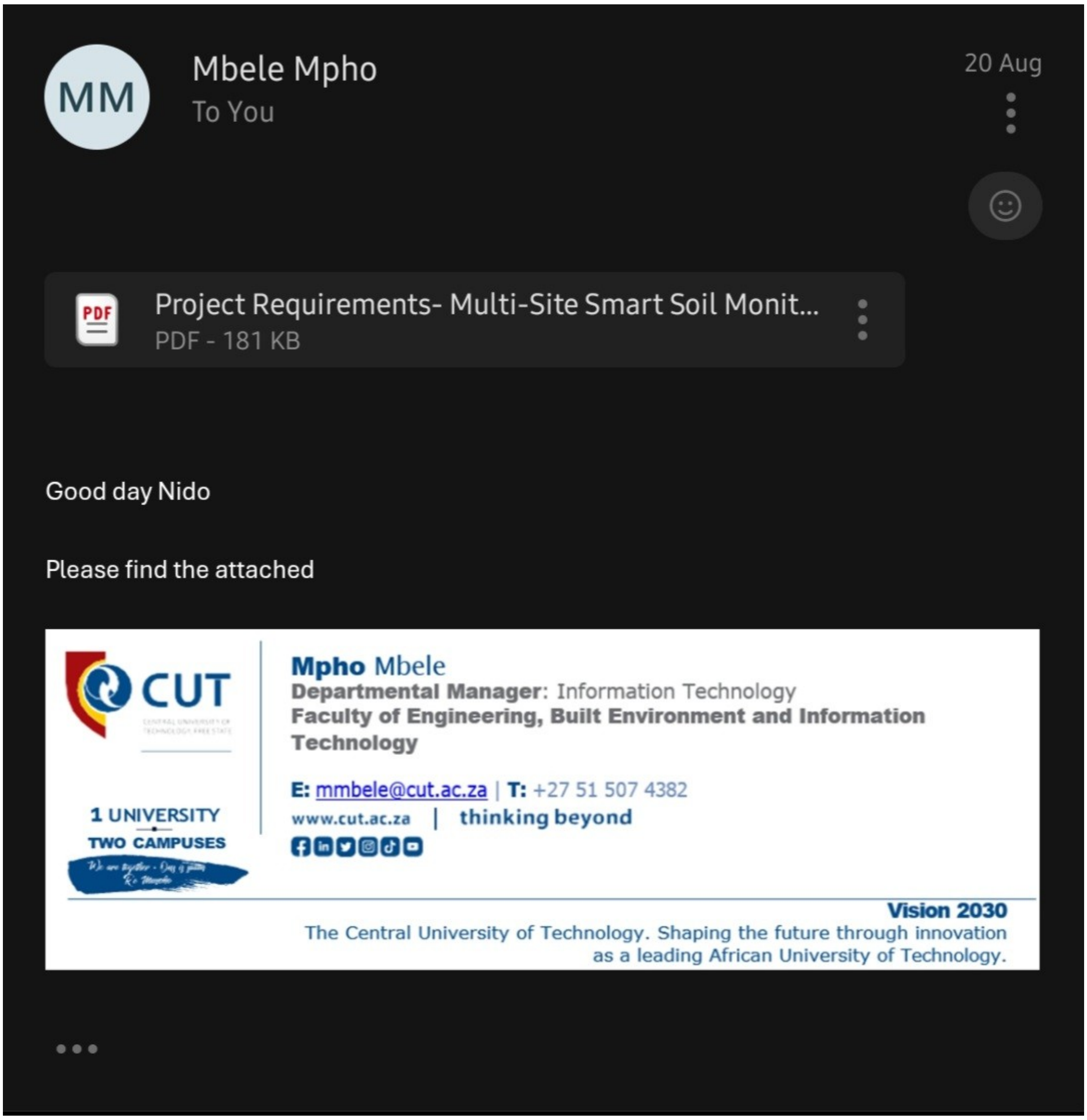
Appendix E — Engagement Record

E.1 Engagement Log

#	Date	Type	With	Subject	Evidence
1	19 Aug 2026	Consultation meeting	Stakeholder	Scope and architecture briefing	Recording and transcript
2	20 Aug 2026	Email received	Ms Mpho Mbele	Project requirements document supplied	Email, screenshot
3	26 Aug 2026	Email sent	Ms Mpho Mbele	Fifteen follow-up questions issued	Email, screenshot
4	27 Aug 2026	Email received	Ms Mpho Mbele	Written responses returned	Email, screenshot
5	30 Aug 2026	Email sent	Ms Mpho Mbele	Findings summary and specification issued	Email, screenshot
6	3 Sep 2026	Email received	Ms Goitsemang Dikane	Findings confirmed and approved	Email, screenshot

E.2 Correspondence Record

Screen captures of the email exchange, in date order. Each confirms the sender, the recipient, the date and the attachment. The consultation meeting that preceded this exchange is recorded in E.3.



Mbele Mpho
To You
20 Aug

Project Requirements- Multi-Site Smart Soil Monit...
PDF - 181 KB

Good day Nido

Please find the attached

CUT
1 UNIVERSITY
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Mpho Mbele
Departmental Manager: Information Technology
Faculty of Engineering, Built Environment and Information Technology

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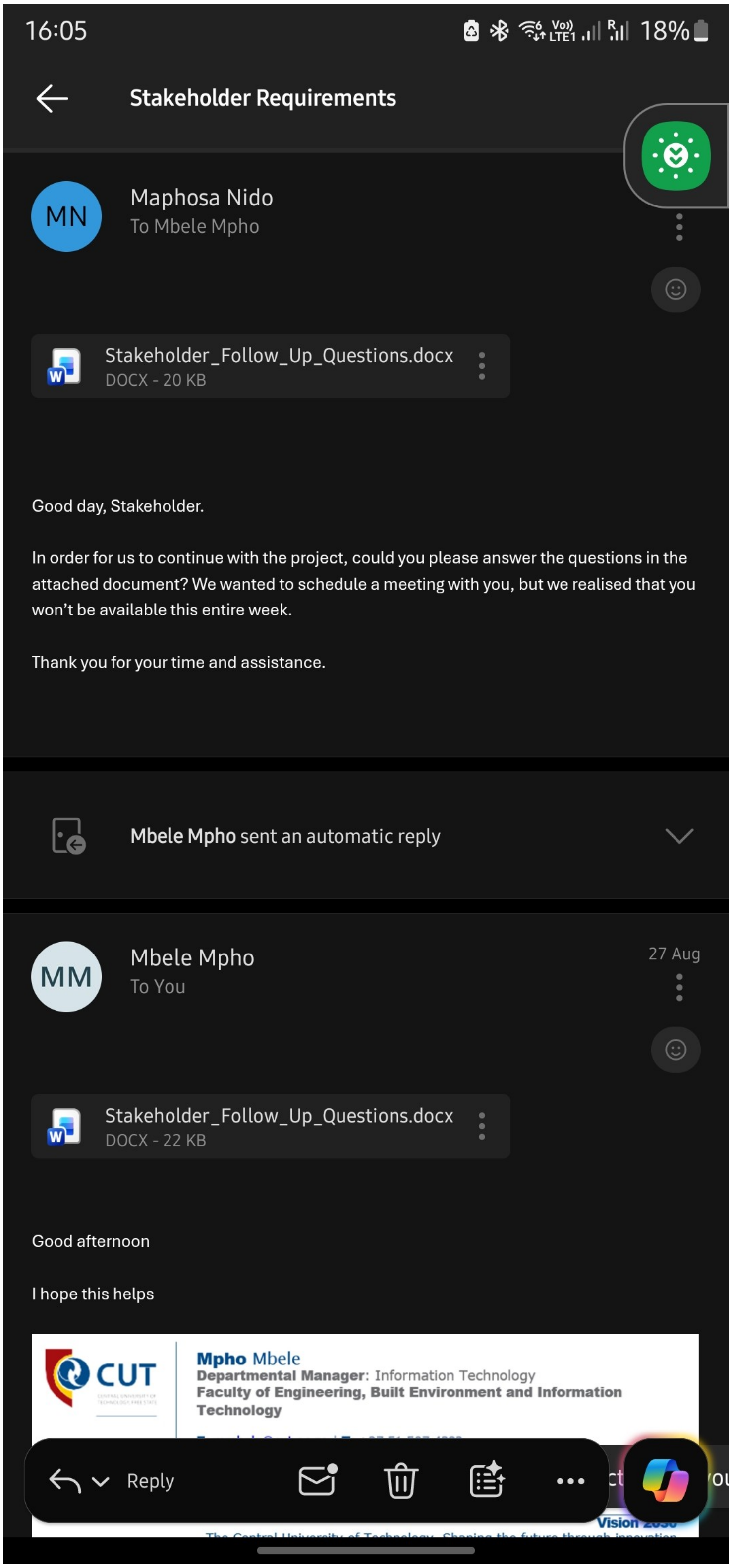
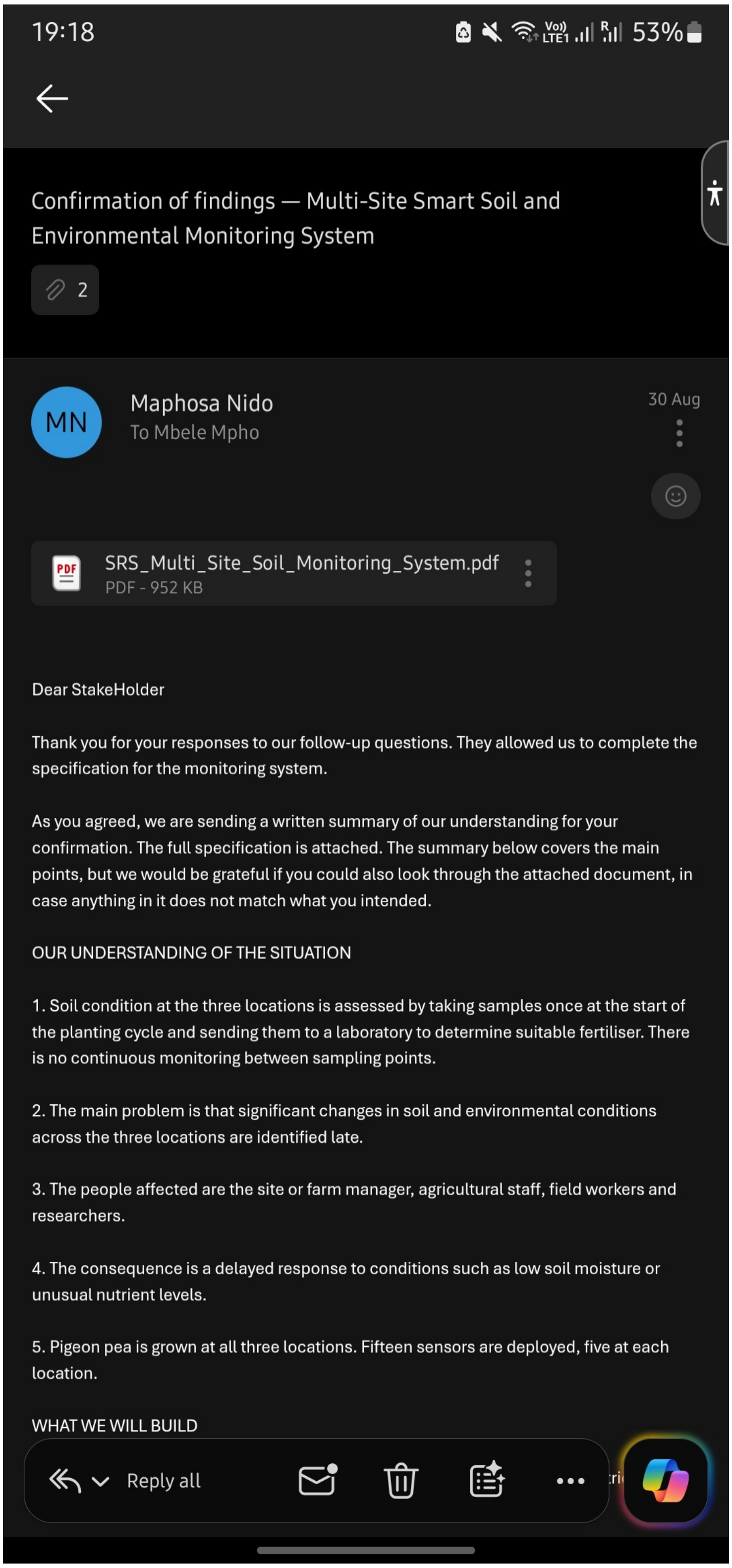
20 August 2026 — Ms Mpho Mbele to Nido Maphosa

Project requirements document supplied. Evidences Appendix C.



26 and 27 August 2026 — questions issued and answered

Sent by Nido Maphosa; responses returned by Ms Mbele. Evidences Appendices A and B.



19:18

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Confirmation of findings — Multi-Site Smart Soil and Environmental Monitoring System

2

MN

Maphosa Nido

To Mbele Mpho

30 Aug

SRS_Multi_Site_Soil_Monitoring_System.pdf

PDF - 952 KB

Dear StakeHolder

Thank you for your responses to our follow-up questions. They allowed us to complete the specification for the monitoring system.

As you agreed, we are sending a written summary of our understanding for your confirmation. The full specification is attached. The summary below covers the main points, but we would be grateful if you could also look through the attached document, in case anything in it does not match what you intended.

OUR UNDERSTANDING OF THE SITUATION

1. Soil condition at the three locations is assessed by taking samples once at the start of the planting cycle and sending them to a laboratory to determine suitable fertiliser. There is no continuous monitoring between sampling points.

2. The main problem is that significant changes in soil and environmental conditions across the three locations are identified late.

3. The people affected are the site or farm manager, agricultural staff, field workers and researchers.

4. The consequence is a delayed response to conditions such as low soil moisture or unusual nutrient levels.

5. Pigeon pea is grown at all three locations. Fifteen sensors are deployed, five at each location.

WHAT WE WILL BUILD

Reply all

30 August 2026 — Nido Maphosa to Ms Mpho Mbele

Findings summary issued with the specification attached, for confirmation.

11:56 5%

Confirmation of findings — Multi-Site Smart Soil and En...

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...

DG Dikane Goitsemang
To Mbele Mpho, You
Yesterday

Dear Me Mbele

Kindly note that I have gone through the document and have realized that some of the minor corrections can be amended while the process unfolds. The calibration and setting of range measurements can also be amended later on once the platform has been established. We shall revisit the pipeline and amend where necessary, for now I believe the document can be used as it meets the entry requirements as the end user.

Warm regards

CUT
Goitsemang Dikane
Plant Sciences: *nGAP*
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20 YEARS OF THINKING BEYOND

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Reply all

3 September 2026 — Ms Goitsemang Dikane to Ms Mbele and Nido Maphosa

Findings confirmed and requirements approved. Evidences Appendix D.

E.3 Consultation Meeting, 19 August 2026

Subject	IoT soil-monitoring platform — scope, architecture and timeline
Present	The stakeholder, and all eight members of the Group O delivery team. One member spoke on behalf of the group during the recording.
Record	Two audio recordings, transcribed. Full transcript held with the project records.
Note	Speaker attribution in the transcript is inferred from turn-taking; no speaker names were captured in the audio.

What the meeting established

#	Established at the meeting	Where it appears
1	Sensor hardware is handled by a separate party. The delivery team does not work on the hardware.	§5.3
2	Readings are transmitted by Wi-Fi and staged in Firebase; the team's role is to bring that data into Supabase.	§8.3
3	Where sensor data is not yet available, the team is to work with representative data in the database.	§8.3, C01
4	Three locations, five sensors at each, fifteen in total.	§2.1, AC3, AC4
5	The database must link locations to sensors, and readings must trace to a named sensor at a named location.	§16
6	Device health is tracked separately. Solar panels and batteries are in use; a failing battery must be flagged.	§10.6
7	Web is the primary home for visualisation; mobile focuses on device and battery health.	§9
8	Readings must be interpreted, not merely displayed.	§15, NFR17
9	Thresholds are to be researched from South African sources; the team may set its own with a stated basis.	§15, §15.1
10	Thresholds are crop-specific; no single national threshold applies to all crops.	§15
11	A threshold breach must raise an alert. Push notification preferred, email secondary.	§10.8, C09
12	Data is fetched at intervals by a scheduled job, at approximately thirty-minute intervals.	§8.3, FR75
13	Climate temperature per location, beyond soil temperature, to be incorporated for context.	See E.4
14	Indicative timeline: Phase 1 early September, design end September, delivery end October.	§17.4

E.3.1 Verbatim extract — scope and data

Stakeholder

> “The data is transmitted by Wi-Fi from the sensor to the database, so there's a middleman — you guys don't really have to worry about the hardware parts. At the moment we're going to read data into Firebase, so your role will just be to take that data from Firebase into Supabase.”

Stakeholder

> “Even if the data isn't ready, we'll just check the type of data that the sensors are supposed to read. So you just have dummy data on your database for those sensors.”

Stakeholder

> “We'll be using three locations. We have fifteen sensors, so for each location we'll put five sensors. In your database you must have locations, and then you need to link locations with the sensors.”

E.3.2 Verbatim extract — interpretation and thresholds

Stakeholder

> “We're not going to leave you with just the raw data — you have to tell us the information and analyse what the data set is saying. So say, for example, pH is 6 — you guys must go and find out what pH 6 means.”

Delivery team

> “The thresholds — will we have to determine them ourselves, or will you provide them?”

Stakeholder

> “You have to go and find out if they exist. If there are no thresholds, then you can come up with your own, just to have something — I can't have you tweaking something with no basis.”

Stakeholder

> “Critical threshold tests for NPK depend on the specific crop type. Other than that, they don't have a general threshold here — it depends on the crop type.”

E.3.3 Verbatim extract — alerting and scheduled collection

Stakeholder

> “Your system must also give alerts. If a threshold is exceeded, then you give out an alert, or maybe send a notification.”

Delivery team

> “Should the notification be from the app, or an SMS?”

Stakeholder

> “From the app. Push notifications are fine, or you can send an email — there's an option for emails. But I think a push notification will do.”

Stakeholder

> “Even though we have solar panels, we're also going to use batteries — so if those batteries are going to die at some point, you need to flag it.”

Stakeholder

> “There's something called a cron job — how to fetch data at intervals, so it fills up your database automatically, instead of only reading once when something happens on the application.”

E.4 Items Raised and Not Yet Closed

Two matters raised in the consultation are recorded here because they are not fully reflected in the specification.

Raised	Position	Status
Climate temperature per location, in addition to soil temperature	The stakeholder undertook to obtain temperature data for each location. This would be an additional external data source beyond the sensors.	Not in the current specification. To be raised before Phase 2 design begins and recorded in the change log with the outcome.
Whether the sensor reports humidity as a measured parameter	The consultation described the sensor as reading temperature, humidity, pH and conductivity alongside nitrogen, phosphorus and potassium. The requirements document, section 3, defines seven parameters and does not list humidity, and the sample reading fields in Appendix B list the same seven.	The specification records seven parameters, following the written requirements document. Recorded as change log entry C04 and to be confirmed with the stakeholder.

E.5 Attendance

Group members present at or involved in each engagement. Also recorded in the Stakeholder Engagement sheet of the project progress tracker.

Member	19 Aug	20 Aug	26 Aug	27 Aug	30 Aug	3 Sep
Nido Maphosa	P	P	P	P	P	P
Caitlin Freeman	P	P	P	P	P	P
Mbasa Tyam	P	P	P	P	P	P
Tiisetso Masita	P	P	P	P	P	P
Machedi Mamello	P	P	P	P	P	P
Ayatolla Pikwa	P	P	P	P	P	P
Katlontle Seiphetlho	P	P	P	P	P	P
Thobani Mjiyakho	P	P	P	P	P	P

P indicates present or involved. All eight members took part in every engagement: the consultation was attended by the group with one member speaking on its behalf, and each item of correspondence was drafted and reviewed by the group before it was sent or acted upon.

E.6 Evidence Held but Not Reproduced

Item	Reason	Where held
Audio recordings of the consultation meeting	Two recordings, 19 August 2026. Not reproduced in a written submission.	Project records
Full meeting transcript	Contains discussion unrelated to the project. The technical content is extracted in E.2.	Project records

Confidentiality

Telephone numbers, identity numbers, credentials and confidential organisational records are excluded from this document and from the source repository. Where a source item contains such information it is redacted before inclusion.